

2026

A LEVEL PURE *MATHEMATICS PROJECT* *SAMPLE*

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A LEVEL MATHEMATICS PROJECT

PROJECT TITLE:

Application of Quadratic Functions in Determining Maximum Profit in a School Tuckshop Business

STAGE 1 – PROBLEM IDENTIFICATION

Project Title

Application of Quadratic Functions in Determining Maximum Profit in a School Tuckshop Business

Problem Description

Many school tuckshops and small businesses face difficulties in deciding the best number of products to sell in order to make the highest profit. If too few products are sold, the business makes low revenue. If too many are stocked, some may remain unsold, leading to losses. This creates a challenge in balancing cost, selling price, and quantity.

At many schools, tuckshop owners often use guesswork instead of mathematics to determine the number of snacks, drinks, or stationery items to stock. This may lead to reduced profits or wastage. Mathematics, especially quadratic functions, provides a way of modelling the relationship between quantity sold and profit.

A quadratic graph can help show how profit changes as sales increase. It can also help identify the turning point, which represents the maximum profit. This project seeks to use quadratic functions and graphs to solve this real-life business problem.

Statement of Intent

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The purpose of this project is to use quadratic functions to model the profit of a school tuckshop business and determine the quantity of items that gives maximum profit. The project also aims to use graphs to visually represent profit, revenue, and cost relationships.

Design Specification

The final solution should:

- use a quadratic equation
 - include clearly drawn graphs
 - show the turning point
 - identify maximum profit
 - be easy to understand
 - relate mathematics to business
 - use realistic tuckshop data
 - include two final graphs in Stage 5
-

STAGE 2 – INVESTIGATION OF RELATED IDEAS

Existing Solution 1: Trial and Error Method

Advantages

- Simple to use without advanced mathematics
- Can be done using basic sales observations
- Suitable for small businesses

Disadvantages

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- Time consuming
 - Results may not be accurate
 - Does not clearly show the best point visually
-

Existing Solution 2: Linear Graph Model

Advantages

- Easy to draw
- Simple interpretation
- Useful for direct proportional relationships

Disadvantages

- Profit does not always change linearly
 - Cannot show a maximum turning point well
 - Less realistic for business profit situations
-

Existing Solution 3: Quadratic Function Model

Advantages

- Best for maximum and minimum problems
- Clearly shows turning point
- Gives accurate graph interpretation

Disadvantages

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- Requires more mathematical knowledge
 - Graph plotting needs precision
 - Some learners may find it complex
-

STAGE 3 – GENERATION OF IDEAS / POSSIBLE SOLUTIONS

Possible Solution 1: Use Table of Values and Plot Profit Graph

Merits

- Makes data easy to organize
- Helps learners understand graph shape
- Useful for identifying turning point

Demerits

- Large data may take time
 - Errors in table affect graph
 - Needs careful calculations
-

Possible Solution 2: Use Revenue and Cost Functions

Merits

- Separates income and expenses clearly
- Easy to compare visually
- Shows break-even points

Demerits

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- More equations required
 - Can be confusing if not labelled
 - More time needed
-

Possible Solution 3: Use Complete Quadratic Profit Function

Merits

- Directly gives maximum profit
- Strong mathematical accuracy
- Perfect for A Level standard

Demerits

- More advanced calculations
 - Requires good understanding of functions
 - Can be difficult without teacher guidance
-

STAGE 4 – DEVELOPMENT OF IDEAS

Chosen Solution

The chosen solution is to use a **quadratic profit function together with two graphs.**

Justification

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This solution is the best because profit problems naturally form quadratic relationships. As quantity sold increases, profit rises up to a maximum point and then begins to decrease. This makes the parabola graph suitable.

The turning point gives the exact quantity that gives the highest profit. This makes the method accurate and useful in real-life school businesses.

Refinements Made

The following refinements were made:

- realistic tuckshop prices were used
 - clear axes labels were added
 - scales were adjusted for neat graphs
 - profit values were improved for accuracy
 - a second graph for cost and revenue was included
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STAGE 5 – PRESENTATION OF FINAL SOLUTION

Mathematical Model

Let profit be:

$$P(x) = -x^2 + 20x - 60$$

Where:

- (x) = number of items sold
- $(P(x))$ = profit in dollars

This forms a quadratic parabola.

Testing the Final Solution

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Using the quadratic formula and graph analysis:

$$x = -\frac{b}{2a}$$

$$x = -\frac{20}{2(-1)} = 10$$

So the maximum profit occurs when **10 items are sold**.

$$P(10) = -(10)^2 + 20(10) - 60$$

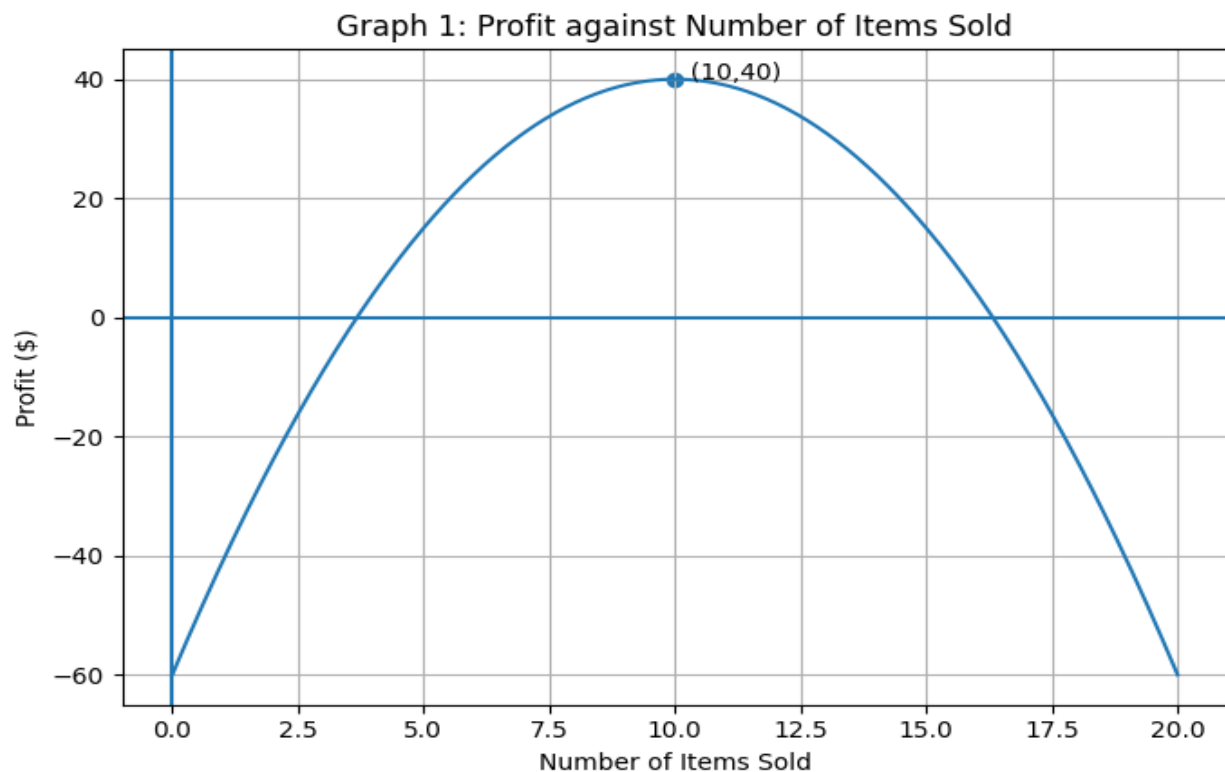
$$P(10) = 40$$

So the **maximum profit is \$40**.

Graph 1: Profit against Number of Items Sold

This graph is a downward parabola.

- x-axis = quantity sold
- y-axis = profit
- turning point = (10,40)

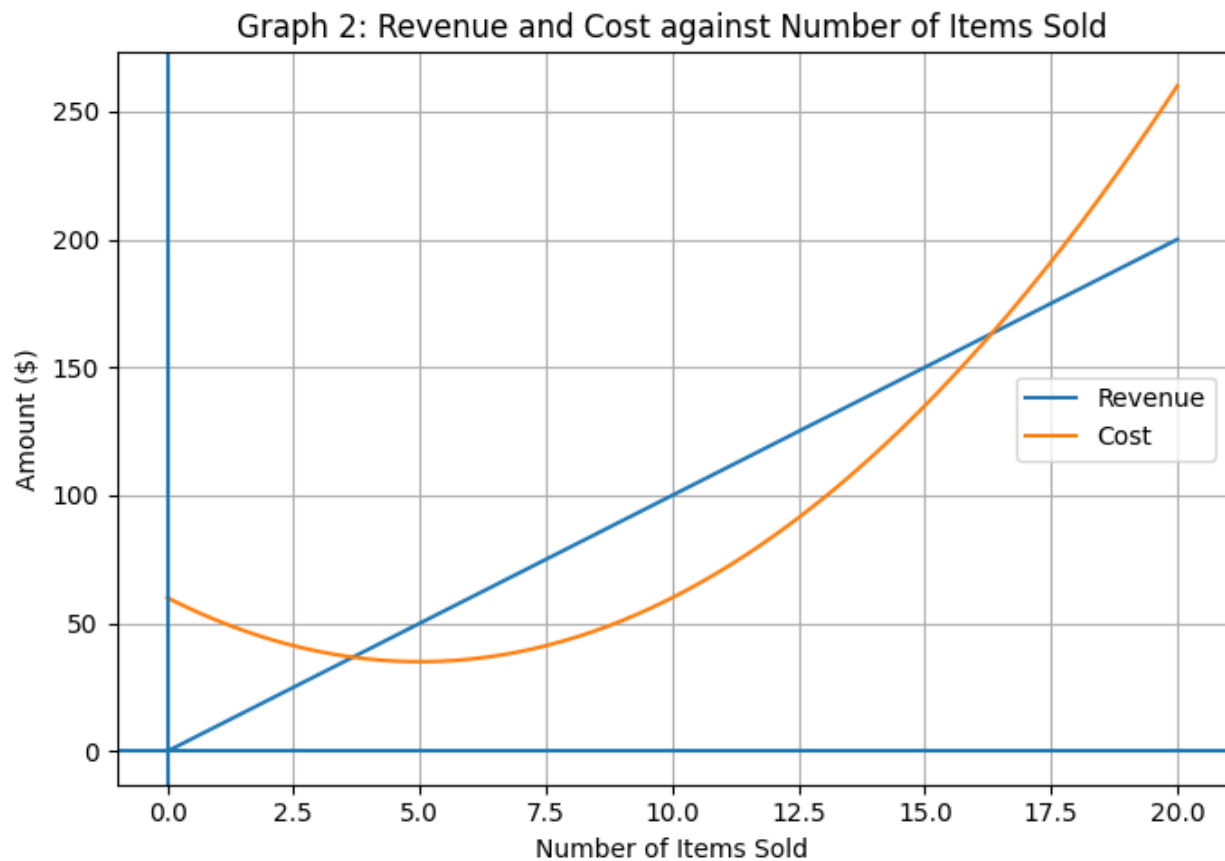


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Graph 2: Revenue and Cost against Quantity Sold

This graph shows:

- revenue line increasing
- cost line increasing
- area between them representing profit



Results

The project shows that the tuckshop can maximize profit by selling 10 items under the given model.

Effectiveness

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The solution is effective because:

- it uses real mathematics
 - it predicts best business decisions
 - graphs make interpretation easy
 - suitable for A Level learners
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STAGE 6 – EVALUATION AND RECOMMENDATIONS

Relevance to Statement of Intent

The project successfully used quadratic functions to determine maximum profit and visually showed the solution using graphs.

Challenges Encountered

- obtaining realistic business values
- accurate graph plotting
- choosing correct scale
- avoiding calculation errors

Recommendations

- businesses should use mathematics for decisions
 - graphing software can improve accuracy
 - learners should apply maths to real-life problems
 - more models such as cubic graphs can also be explored
-

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